World Models for Physical AI: Learning to Predict, Plan, and Act in the Real World

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World models predict environmental evolution; action-conditioned variants model the consequences of specified actions. These models increasingly support simulation, policy learning, planning, and evaluation in physical AI. However, differences in representations, action interfaces, and evaluation protocols obscure when better predictions yield better decisions. This survey reviews data foundations, modeling approaches, and applications in robotics, navigation, and autonomous driving. Drawing on experiments and counterexamples, we examine gaps between recursive accuracy and planning utility, evaluator agreement and policy-training outcomes, and predictive information and execution-time correction. We organize evidence across five uses, identify unresolved attribution questions, and discuss requirements for reliable, action-grounded world modeling.

CCS Concepts: • **Computing methodologies** → **Robotics**; *Reinforcement learning*.

Additional Key Words and Phrases: world models, Physical AI, robot learning, predictive representations, vision-language-action models, model-based control

1 Foundations and Scope

1.1 Introduction

World models learn to predict how an environment evolves, from future observations and latent states to physical interactions and action consequences. In Physical AI, these predictions support simulation, policy learning, action selection, and behavioral evaluation. Yet predictive accuracy does not necessarily translate into better downstream decisions: the benefit depends on what is predicted and how that information enters learning or control.

Predictions can shape a policy during training or guide actions during execution. *World Models* trains a controller in a learned environment [22], whereas PlaNet uses predicted latent trajectories to select actions online [24]. The action interface also matters. OpenVLA maps observations and instructions directly to robot actions without an explicit interface for querying alternative outcomes [41]; V-JEPA 2 uses an action-conditioned predictor to support planning over candidate actions [2]. In autonomous driving, Drive-WM forecasts supplied maneuver candidates, whereas OccWorld jointly predicts occupancy and the ego trajectory [84, 107]. Together, these examples show why the decision interface cannot be inferred from the output modality alone.

This survey examines how world-model predictions contribute to learning and decision-making, and what experimental evidence supports their downstream utility. We compare their use in learned simulation, policy learning, policy evaluation, planning, and execution-time correction. Three cross-study findings guide this survey. First, recursive predictive accuracy and planning utility are distinct validation targets, as illustrated by GNS and ContactWorld [71, 104]. Second, evaluator agreement does not necessarily translate into policy-training gains: DexTouch-WM and IWS show that the utility of generated experience depends on the intended learner and training protocol [67, 85]. Third, the value of execution-time prediction depends on when information becomes available and which actions remain editable, as illustrated by FutureRTC and TacPAC [38, 59].

These findings motivate related attribution questions concerning candidate availability versus selection, predictive representation versus goal scoring, and anticipation versus execution-time feedback. Section 4.4 develops corresponding diagnostic controls while distinguishing proposed experiments from reported evidence.

Section 1 establishes the analytical framework and review scope. Section 2 examines supervision, predictive representations, and generalization; Section 3 compares how predictions enter learning

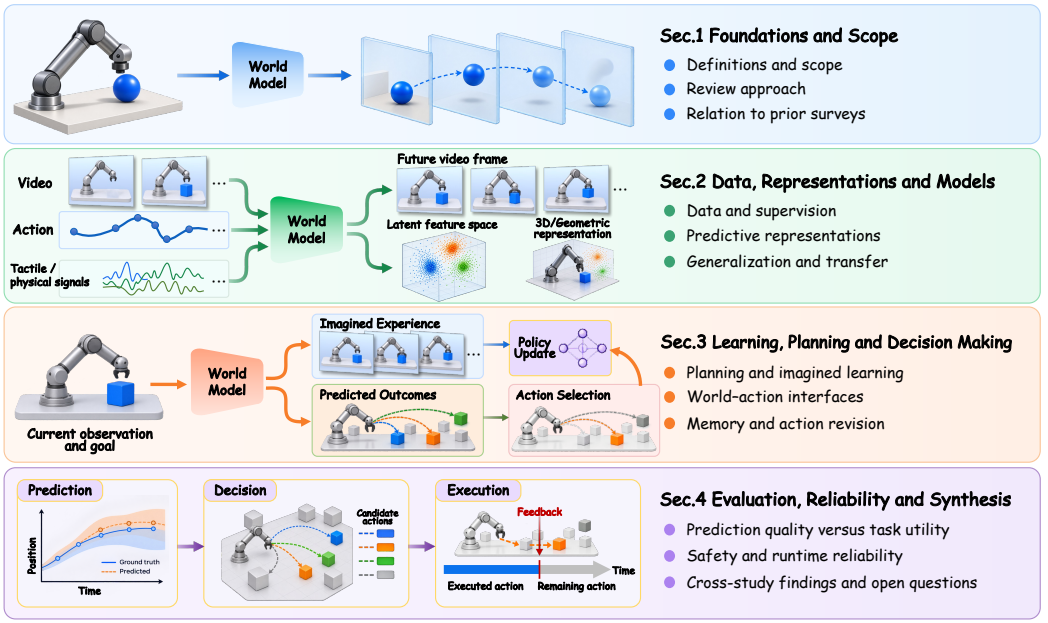


Fig. 1. Organization of the four-part survey. Scenes and curves are conceptual illustrations, not reported experiments; prediction quality, decision utility, and execution reliability require distinct evaluation.

and action; and Section 4 evaluates downstream utility, safety, execution reliability, and remaining attribution questions. Figure 1 outlines this structure.

1.2 Analytical Framework: Prediction, Action, and Use

A world model can be situated along three descriptive axes before its performance is assessed. *Prediction target* identifies the future quantity: pixels or video, latent state, objects and geometry, or physical and multimodal variables. *Action query* asks whether the predictor is passive, responds to supplied candidate commands, generates a desired future, or jointly produces actions and futures. *Prediction consumer* identifies where the forecast enters computation: simulation, policy learning, policy evaluation, planning and selection, or execution-time monitoring and correction. Together, these axes describe interfaces that can recur across model families, building on distinctions in existing robot-world-model surveys [28, 94].

For each located design, our evidence analysis adds three audit fields: the *operation actually changed*, the *measured endpoint*, and the *attribution boundary*. These fields link each design to the evidence for its claimed use, keeping descriptive coordinates distinct from experimentally supported utility. For example, a desired future is not an independently queried consequence of a specified command, and jointly generated actions do not imply counterfactual evaluation of alternatives. State acquisition matters too: GNS predicts dynamics from supplied particle states, whereas ParticleFormer infers point-cloud states from stereo observations before forecasting them for MPPI-based control [31, 71].

The five predictive uses differ in the computation they change and the evidence needed to evaluate them. *Learned simulation* advances predicted states and is assessed through rollout accuracy or behavior in simulated interactions. *Policy learning* uses imagined transitions, returns, or demonstrations to update policy parameters; the relevant endpoint is the trained policy's task performance. *Policy evaluation* estimates a policy's outcomes or relative quality and its agreement

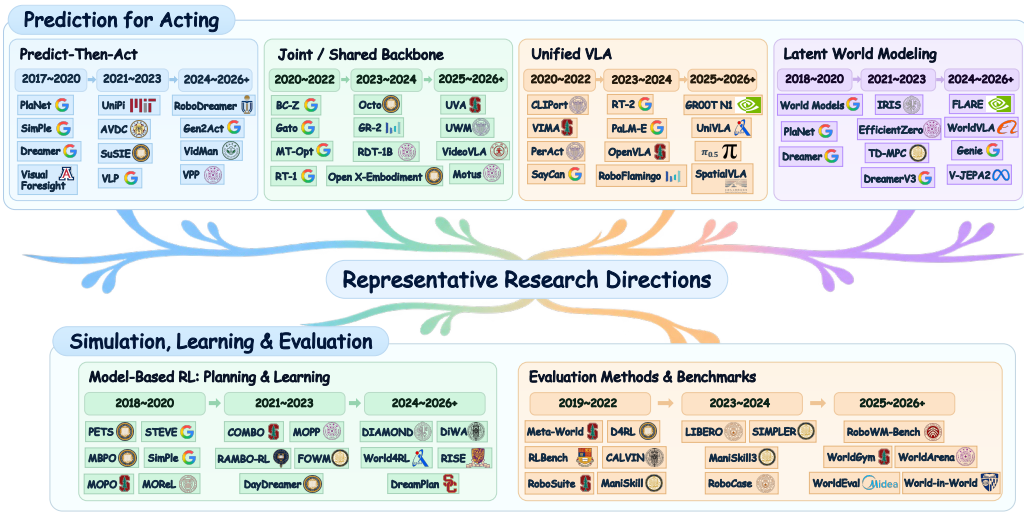


Fig. 2. Illustrative map of prediction for acting, model-based planning and learning, and evaluation resources. Open X-Embodiment is a data resource; LIBERO, CALVIN, RLBench, and ManiSkill are benchmarks or simulation resources. Positions indicate thematic relationships; preprint and publication years may differ.

or calibration must be tested against outcomes observed independently of the evaluator. *Planning and action selection* uses predictions to choose among actions and is assessed by task outcomes under a specified protocol; offline ranking proxies are not physical execution results. *Execution-time monitoring and correction* uses predicted risks or discrepancies to trigger or revise behavior; detection, action revision, and successful recovery are distinct endpoints. One model may serve several uses, but evidence from one does not establish benefit in another.

A reported improvement must also be interpreted at the level of the actual intervention. Changing the encoder, predictive model, goal score, training data, controller, or search procedure together supports a system-level comparison, not necessarily a predictor-only effect. Data-collection and planning budgets, candidate-query counts, feedback timing, and the statistical unit may likewise differ; Moerland et al. already distinguish the budgets for real data collection and planning [62]. We therefore record what was changed and held fixed, which endpoint was measured, and which explanation remains unresolved. For example, Yuan et al. observe alternative outcomes in a fixed same-state candidate bank, whereas DreamAvoid records executed failures without observing rejected branches [15, 96]. Section 4.4 develops the remaining conditional tests.

1.3 Working Definitions and Inclusion Boundaries

We use *world model* for a learned predictor of environmental evolution, including future observations, latent states, object relations, physical quantities, or action consequences. An *action-conditioned* model predicts consequences under specified actions or interventions; passive models need not expose executable-action queries. Joint world–action systems fall within this definition when they predict environmental change, not merely because they generate commands.

Prediction, policy, and control are computational roles, not necessarily separate networks. Predictors estimate future states or consequences, policies select actions, and controllers realize motion. A system may combine these roles, but action generation alone does not establish consequence prediction.

Our direct cases examine learned predictions used for simulation, policy learning or evaluation, planning, and execution-time monitoring or correction. Datasets, analytical simulators, standalone encoders, sensors, and reactive or direct-action policies serve as resources or boundary comparisons unless they expose a relevant predictive mechanism. Training-only future prediction can contribute to policy learning without implying deployment-time prediction queries.

1.4 Review Scope and Evidence Selection

This survey adopts a structured critical narrative approach to examining world models for Physical AI. We review literature spanning predictive representations, action-conditioned modeling, simulation, policy learning, planning, evaluation, and execution-time correction. Our focus is on how learned predictions support downstream operations and what evidence establishes their practical utility.

Our mechanism-focused selection prioritizes studies with original experiments, controlled contrasts, or informative counterexamples that illuminate downstream uses. This purposive, non-exhaustive selection targets mechanism-level attribution rather than field-wide prevalence estimates.

For each comparison, we examine the experimental setting, the components changed or held fixed, the measured outcomes, and the limits of causal attribution. We distinguish controlled interventions from coupled-system comparisons and identify unresolved explanations where the available evidence is insufficient.

1.5 From Learned Dynamics to Predictive Robot Systems

Learned dynamics have long supported different forms of decision-making. Dyna integrates model learning with incremental planning and reactive execution [78]. PETS uses probabilistic dynamics ensembles and trajectory sampling for online action selection [12], while PlaNet plans over trajectories in learned latent space [24]. Model-based prediction also supports policy learning: MBPO adds short model-generated rollouts branched from real experience to off-policy updates [35], while Dreamer trains an actor using returns imagined in latent space [23]. These approaches differ in whether predicted transitions inform the current action or update a policy for later use.

Physical robot learning extends these uses without making them equivalent. DayDreamer trains policies from imagined experience while learning directly from real robot interactions, without an external simulator [87]. DreamZero takes a different route, jointly generating future video and actions from a pretrained video-model backbone [95]. Joint generation supports action production, but does not by itself establish an interface for predicting the consequences of arbitrary supplied actions. Figure 2 situates these methods alongside related prediction-for-acting research and simulator/data/benchmark resources; the map is illustrative, not a uniform list of world-model consumers.

The distinction can be expressed through the model's query. Given an observation–action history h_t, a supplied action sequence $a_{t:t+H-1}$, and an optional goal g, an *action-conditioned predictor* models

$$p_\theta(y_{t+1:t+H} \mid h_t, a_{t:t+H-1}, g), \tag{1}$$

where y denotes future observations, latent states, or physical variables. By contrast, an action-generating policy models $\pi_\phi(a_{t:t+H-1} \mid h_t, g)$, while a joint world–action generator models the distribution of future states and actions together. Passive prediction may omit actions altogether. These expressions identify distinct information interfaces, not required network architectures or evidence of downstream benefit.

1.6 Relation to Recent Surveys and Review Scope

Existing surveys characterize complementary aspects of predictive representations, world–action architectures, and evaluation. Hou et al. distinguish training-time world modeling from optional inference-time generation [28]. Lu et al. classify joint/IDM world–action architectures, evaluation protocols, and runtime coupling (Secs. III, V and VI-F) [58], and Shen et al. examine deployment regimes and runtime budgets (Secs. 4.5 and 7.1) [74]. Kirchner et al. discuss physical structure, model bias, and closed-loop control (Secs. 3.6 and 4.1) [43]. Wang et al. separate predictive gains from pretraining, search, and reward scoring (Sec. 9.2) [82]. Yao et al. organize prediction consumers and matched counterfactual controls (Secs. 6 and 9.2; Table 3) [94]; Jena et al. formulate capability-specific evaluation protocols (Secs. 4–5; Table 9) [36]. Together, these surveys provide the analytical foundation for comparing prediction with downstream use.

Our contribution connects primary-study contrasts to three consumer-specific validation decisions. GNS and ContactWorld motivate testing recursive stability alongside goal-scored planning; ContactWorld's predictive representation and goal geometry remain coupled [71, 104]. DexTouch-WM and IWS motivate testing evaluator agreement and generated-data utility for the intended learner [67, 85]. FutureRTC and TacPAC motivate checking when predictive information becomes available relative to executable actions [38, 59]. Section 4.1 presents the experimental controls and task-level counterexamples, and Section 4.3 examines the runtime mechanisms.

Two further contrasts connect these decisions to the action interface. Fast-WAM's fixed-inference video-loss control tests training supervision, while LaRA-VLA's separate supervision and inference comparisons test a joint reasoning/visual context bundle [4, 97]. Their interventions distinguish predictive supervision during training from access to intermediate context during action generation; LaRA's controls leave the visual and reasoning contributions coupled. The original study *Beyond Visual Quality* establishes an outcome-oracle selection gap within a fixed action bank [96]. Comparing that observed bank with DreamAvoid's unobserved rejected outcomes identifies the evidence needed to distinguish an unsuccessful selection from the absence of a viable alternative [15]. These primary-study contrasts complement the seven surveys above by relating supervision to deployment-time readout, evaluation agreement to learner utility, and observed candidate outcomes to selection. Section 4.4 develops the resulting diagnostic tests.

2 Data, Representations, and Predictive Models

Predicting observed motion and predicting a robot command's consequence require different information. This chapter connects data interfaces, predictive representations, and adaptation across scenes and bodies through the computation consuming each prediction. The comparison follows information available for simulation, learning, and action selection rather than representation richness alone. Section 3 develops those computations; Section 4 examines their utility. Table 1 records the supervision interfaces, query modes and downstream tests summarized across this chapter.

Figure 3 illustrates the connections among data, predictive representations, downstream uses, and generalization considered in this survey.

2.1 Data: From Observed Motion to Executed Actions

The data bottleneck is connecting observed variation to the intervention a model must predict. Passive video, synchronized commands, physical-state labels, and recovery annotations supply different supervision for action queries, policy updates, and interaction outcomes. Their value depends on the operation they enable, not trajectory count alone.

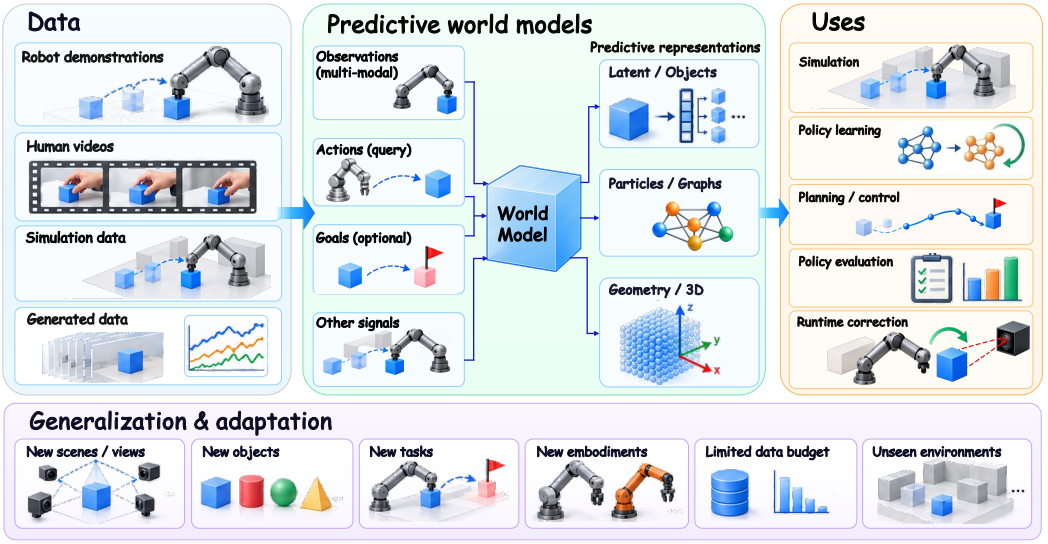


Fig. 3. Overview of data sources, predictive representations, downstream uses, and generalization.

Broad video experience needs a bridge to executable commands. V-JEPA 2 freezes an action-free pretrained encoder and trains a predictor on synchronized robot observations and base-frame end-effector changes [2]. DreamDojo instead learns continuous latent-action proxies from human video, then replaces them with robot actions during target-domain post-training [17]. The first bridge establishes command conditioning after visual representation learning; the second changes the condition of an already pretrained predictor. Both need robot trajectories; V-JEPA 2's unlabeled data omit task, reward, and success labels, not action signals. DreamGen reverses the interface: after embodiment adaptation, instruction-conditioned generated trajectories receive inferred latent-action labels or inverse-dynamics action chunks for offline policy training [34]. These recovered actions are inferred labels for generated video, not measurements of commands executed by a robot; errors in the generated motion or contact can propagate into inverse-dynamics labels. Demonstration generation therefore supplies actions inferred from futures, whereas a supplied-action predictor answers queries about commands already specified. Plausible generated behavior and an accurate response to a chosen command are different supervision products, even when both reuse a video prior.

Synchronized corpora establish that command–observation link without erasing control conventions. DROID broadens scenes on shared hardware; Open X-Embodiment pools bodies with coarsely aligned end-effector actions but retains frame and control differences [39, 64]. Their policy co-training comparisons assess data mixtures. Reusing DROID to fit V-JEPA 2's predictor instead assesses action-conditioned model learning [2]. A shared storage format does not itself align executable command meanings: reference frames, rotation and gripper conventions, timestamps and control frequencies, action-chunk duration, and embodiment-specific control protocols remain part of the supervision. Otherwise, pooling trajectories can change which transition a training pair describes rather than simply broadening the situations covered.

Physical labels and interactive environments offer complementary access. PhysInOne supplies rendered evolution with depth, trajectories, materials, and text under asset-disjoint partitions, supporting prediction, property estimation, and resimulation, without described robot-command trajectories [111]. GRUtopia instead exposes scenes and embodiment-specific control APIs through

which agents generate interactions [83]. Annotation reveals consequences; an environment permits new queries. In the former case, the predictor learns from recorded physical variables; in the latter, a controller can request another interaction. These resource roles do not themselves measure a learned predictor's control benefit. PhysInOne's SVD–LoRA comparison also improves FVD while lowering Physical Motion Fidelity, a Fourier-energy motion measure. Video-distribution fit and physical-motion agreement thus assess different properties, neither being a control result.

Contact and recovery require more than synchronized motion. DexTouch-WM aligns tactile layouts, coordinates, vision/touch streams, and kinematic retargeting to transfer human interaction supervision into robot-domain prediction [67]. This alignment makes contact consequences available beyond the human recording, while contact-overlap assessment distinguishes contact regions from mostly non-contact reconstruction. A shared tactile layout and coordinate calibration establish correspondence between human and robot sensory consequences, while retargeting addresses the motion interface. Its prediction, evaluation, and policy-training experiments are compared in Section 4.1.

Failure trajectories provide two kinds of supervision depending on the consumer: improving consequence prediction or supplying explicit recovery actions. VLAW uses both successful and failed real rollouts to fit its action-conditioned video model, while policy fitting uses real successes and reward-filtered synthetic successes [21]. The observed decrease in optimistic predictions and increase in missed successes reflect an augmented recipe that changes data volume and outcome balance together. REBOOT instead timestamps failure, correction and completion phases, exposing demonstrated corrective actions and valid continuation states [63]. Outcome-level failure examples inform predicted consequences, whereas phase-annotated segments supervise recovery behavior.

2.2 Visual Prediction and Learned Simulation

The first design choice is whether a video specifies what to achieve or answers what a supplied action will do. UniPi generates a desired trajectory from an initial image and task text, then independently trained inverse dynamics recovers commands [13]. AVID instead conditions generation on supplied actions [70]. The former decodes commands from a desired future, whereas the latter tests whether generated motion is consistent with a supplied command. Neither inverse-dynamics decoding nor visual action consistency alone establishes physical feasibility; that requires evidence from executed motion and interaction. Similar-looking videos can occupy opposite sides of this interface.

Attaching commands to a pretrained generator depends on both data and model access. AVID's action-conditioned adapter reads noisy video, the initial image, the base model's noise prediction, and the diffusion step; a learned mask blends base and adapter predictions for the denoising loss without updating unavailable base parameters [70]. The mask determines which parts of the prior's prediction are retained or replaced, so target actions need not be recoverable from that visual prior alone. DreamDojo learns the robot interface through post-training after human-video proxy-action pretraining [17]. OSCAR instead conditions video on calibrated projected human/robot skeleton trajectories [89]. Action-labelled target videos, target-robot fitting, and calibrated kinematics are different bridges from visual priors to motion, not a common cross-body actuator space.

Continuing interaction requires using imperfect predictions as later context. UniSim carries finite observation-frame history across segments and supplies a new action, omitting earlier actions from that history [92]. Its common input representation accepts language and processed low-level controls with source-dependent meanings. Interactive World Simulator retains bounded latent and action histories; its image encoder and consistency-model decoder are learned and frozen before latent-dynamics training [85]. At inference, the predicted latent is appended to context and old latents beyond the window are discarded; the decoder then renders the new observation. Training-context perturbations address the mismatch between observed and self-generated inputs.

Table 1. Representative supervision and model interfaces (§2). Symbols encode an *explicitly documented interface*, not model accuracy or task success.

Study / resource	Data or predictive target	A	G	S	Downstream use / test
Data and environments (not learned predictive queries)					
DROID [39]	Robot demonstrations	–	–	–	Policy transfer
OXE [64]	Cross-body commands	–	–	–	Co-training
PhysInOne [111]	Physics-labeled video	–	–	–	FVD / PMF
REBOOT [63]	Failure / correction phases	–	–	–	Recovery data
GRUtopia [83]	Control API / scenes	–	–	–	Agent task tests
LIBERO / CALVIN [52, 60]	Task benchmark	–	–	–	Task / chain success
Visual and latent prediction					
V-JEPA 2 [2]	Robot-conditioned latent	✓	–	–	Planning / probes
UniPi [13]	Desired future video	×	✓	–	Inverse dynamics
AVID [70]	Action-conditioned video	✓	–	–	Frame / AER
UniSim [92]	Interactive video	✓	–	–	Video / policy
IWS [85]	Latent / action histories	✓	–	–	Evaluation / training
DreamDojo [17]	Robot-adapted video	✓	–	–	Adaptation tests
OSCAR [89]	Projected skeleton video	×	–	✓	Evaluation agreement
Dreamer [23]	Latent states / rewards	✓	–	–	Policy return
Structured / geometry / contact prediction					
C-SWM [42]	Object successor	✓	–	✓	Successor ranking
SlotFormer [88]	Post-intervention slots	×	–	✓	PHYRE AUCCESS
GNS [71]	Particle acceleration	×	–	✓	Rollout error
MeshGraphNets [66]	Mesh/node dynamics	×	–	✓	Rollout error
TesserAct [106]	Future RGB-DN / 3D	×	✓	✓	RLBench 7+/2–
OmniVTA [108]	Tactile deformation	×	–	✓	Action refinement
ContactWorld [104]	Contact-state latent	✓	–	✓	CEM planning
DexTouch-WM [67]	Vision/touch dynamics	✓	–	✓	Evaluation / training

Note. A = supplied-action consequence queries; G = goal-conditioned future generation; S = explicit object/geometry/particle/contact structure. ✓ = documented; × = not exposed in the cited setup; – = not applicable or not established.

Both models recurse, but differ in retained history and preparation for recursive error. Separating image reconstruction from latent dynamics also makes explicit which learned components stay fixed while the temporal predictor is fitted.

Their prediction tests probe different links in that computation. UniSim's Ego4D ablation favors recent over distant frames on video-distribution and appearance measures, whereas AVID compares frame fidelity with regressor-based action consistency [70, 92]. Interactive World Simulator compares generated rollouts with recorded-action references; longer interactive demonstrations show continued generation rather than reference-matched accuracy throughout [85]. Context usefulness, commanded-motion consistency, and accumulated prediction error consequently need distinct measurements. None of these individual tests simultaneously measures command realization and accumulated error under a policy's changing actions.

Returning generated observations to an agent changes the consumer, even when the predictor stays the same. In Interactive World Simulator, teleoperators supply commands to collect synthetic demonstrations, whereas policy evaluation feeds generated observations back to a policy [85]. The first interface creates training examples; the second lets the tested policy choose its subsequent inputs to the simulator. Section 4.1 compares their empirical outcomes and protocols. Neither use can be reduced to the fidelity of an independently sampled video.

Interactive visual simulation couples command-conditioned prediction with observation feedback. Command consistency tests whether the generator follows a supplied intervention; recursive-error tests assess how predicted observations affect later predictions. Both inform simulator selection, while evaluation of a reacting policy additionally requires comparison with independently observed policy outcomes. Action decoding and consumer-specific utility are examined in Sections 3.5 and 4.1.

2.3 Latent and Structured Predictive Representations

A representation matters through the distinctions its consumer needs: derivatives for an integrator, attainable geometric change for inverse dynamics, or anticipated deformation for contact control. Latent compression, object structure, geometry, and touch can overlap. The relevant path is therefore state acquisition, predicted evolution, and the operation receiving that prediction, not representation richness alone.

A compact encoder is not yet a queryable transition. LeVJEPA aligns global/local views with a shared encoder and distribution regularization; block-causal attention restricts temporal access without adding an action-conditioned successor model [44]. Matched-pretraining-FLOP probing improves Kinetics-400 but lowers Something-Something-v2 accuracy relative to V-JEPA 2. Dreamer adds observation-updated state estimation and action-conditioned latent/reward transitions [23]. The observation update combines previous state/action with the current observation; an imagined transition omits the corresponding future observation. Its actor/value learner consumes imagined returns without rendering each image. Frozen probing and policy optimization thus test different uses of compression, with transition and reward models supplying the quantities the optimizer actually needs.

Object structure likewise needs an appropriate readout. C-SWM predicts action-conditioned image-derived object embeddings through pairwise messages and contrastive successor matching rather than pixel reconstruction [42]. SlotFormer instead evolves aligned slots from a scene in which the PHYRE object-placement intervention is already rendered; a supervised success readout ranks rollouts [88]. Its within-template PHYRE AUCCESS is 82.0 versus SAVi's 80.7 over ten folds with a 100-attempt/task budget; this tests rollout plus readout, not object factorization alone. Successor ranking and outcome-based action ranking are distinct endpoints.

With supplied physical state, graphs organize the numerical update rather than discover objects. GNS predicts particle accelerations and integrates velocities/positions, rebuilding proximity relations as particles move [71]. MeshGraphNets combines mesh and spatial-contact connectivity, integrating node outputs and optionally remeshing resolution [66]. GNS's fluid, granular, and deformable tests and MeshGraphNets' cloth, structural, and fluid tests start from their respective physical states. Both study state evolution rather than camera-to-state acquisition; one-step fitting and recursive integration impose different demands. Hybrid models retain an analytical backbone: Heiden et al. fit friction alongside analytical normal contact in sim2sim and physical parameters in a real double pendulum [27]. PGRD instead fits base parameters for an RGB-D-derived spring–mass state and adds learned velocity residuals to its position update [65]. Zeroing residuals on held particles preserves gripper boundary conditions and lets MPPI query candidate gripper actions. Its correction changes simulated object evolution, not the policy's command directly.

Visible geometry can instead supply an intermediate condition for action decoding. TesserAct predicts RGB, depth and normals from an initial RGB-DN observation and robot-specific task instruction, reconstructs future visible point clouds, then passes those features with current state and instruction to inverse dynamics [106]. During execution, current state is refreshed against precomputed future keyframes. The coupled representation-and-decoder system outperforms reimplemented UniPi on seven of nine RLBench tasks and loses on two. Its geometry tests use simulator depth in RLBench and estimated depth/normals on real video; the latter tests predicted

geometry, with no corresponding physical-execution endpoint. Unlike GNS and MeshGraphNets, whose integration advances supplied material states, TesserAct uses reconstructed surfaces to condition future-to-action decoding.

Touch separates action generation from action search. OmniVTA conditions visual-tactile prediction on observations and projected historical end-effector positions, not candidate future commands [108]. Current/predicted deformation features and their difference inform slow action chunks; a separately trained fast controller combines anticipated/observed touch and robot state to refine actions. Future vision improves wiping but lowers peeling with little mean benefit and increased latency, so deployment omits that branch while retaining joint prediction during training. The fast controller is trained separately on normal-contact and recovery samples. Its controller-on/off ablation therefore tests a different intervention from supplying another predicted modality to the slow generator.

ContactWorld instead feeds candidate commands to a GRU predictor and uses CEM to minimize terminal distance to a goal latent, executes the first selected action, and replans [104]. Modality configurations share predictor family, objectives, and planner but learn different encoders and goal geometries. Stable prediction and successful planning favor different configurations; cost and capacity controls retain aggregate benefits with mixed task effects. Section 4.1 presents the full comparisons, numerical results, and trial counts. The mechanism links contact information to an action-search objective, while the controls leave its coupling to learned goal geometry open; Section 4.4 proposes a common-metric test.

Finally, a common sensory representation can serve different consumers. DexTouch-WM returns imagined observations to policies during evaluation, but retains recorded action sequences when replacing future observations for training [67]. The evaluator lets policies react to simulated consequences; the training recipe changes sensory examples along supplied actions. Section 4.1 compares these evaluation and training uses; the human-data prediction-scaling experiment is a separate test. The shared interface matrix in Table 1 identifies these consumer paths; the update, readout, decoder, or scorer determines which distinction is useful.

2.4 Generalization Across Scenes and Embodiments

Scene, task, sensor, and body changes stress different interfaces. The comparisons below distinguish three bridges: transferring experience, adapting executable actions, and preserving selected outputs under observation changes. These bridges can coexist, but their evidence concerns different changes and consumers rather than a single generalization score [64, 95].

Experience transfer broadens situations without necessarily changing the executing body. Sim-to-real WAM adapts Cosmos Policy with approximately 800 simulated demonstrations per task and visual domain randomization, reporting 35% mean zero-shot success across four Franka manipulation tasks (10 physical trials per task), without real policy demonstrations or real-data finetuning [86]. This measures the complete transfer recipe—including video pretraining, generated demonstrations, and domain randomization—rather than an isolated contribution of world-model prediction. Scene transfer here retains a physical command interface while changing the observation domain. OXE instead pools real-robot experience across existing interfaces [64]. DreamZero co-trains AgiBot data with unseen-task videos from YAM or humans, improving task progress after additional fitting [95]. Those progress scores concern experience transfer, not execution on an unadapted source body. Additional co-training, rather than a zero-update test, is the link between the new visual experience and the receiving policy.

Task specification and future use create distinct tests within this bridge. Zero-WAM uses human videos to specify held-out configurations on a fixed bimanual Franka and improves over language conditioning after seen-task robot adaptation [110]. Temporal Ratio instead gates guidance using

attention to future versus current latents, improving physical object–receptacle recombination on one YAM [61]. The former changes the task prompt; the latter changes how an existing policy uses predicted context. Comparisons of Zero-WAM with independently trained language- or video-conditioned systems also differ in data, model design or training budget, so they do not isolate the causal advantage of the prompt modality. Guidance costs additional video passes and can amplify an incorrect future, so useful prediction and useful conditioning remain separate requirements.

Action adaptation bridges shared experience to a particular body. OSCAR's calibrated skeleton trajectories share projected motion conditions across humans and robots, but are not hardware commands [89]. DreamZero separately fits its AgiBot model to YAM trajectories for new-body execution [95]. DyPES-VLA retains body-specific realization inside joint training: future-frame supervision shapes shared query states and statically routed experts decode native action chunks [48]. Its future head does not take candidate commands and is skipped at deployment. Removing future supervision or replacing experts with a dense head lowers simulated success; physical deployment includes target-embodiment finetuning. Sharing a prediction condition and fitting an executable policy are different achievements.

Selective invariance addresses sensor/view changes while retaining action semantics. SCVC constrains action, proprioception, and value outputs for same-state camera pairs but lets future images vary with viewpoint [30]. Base-frame end-effector increments, unchanged task semantics, and two task-informative views make that distinction meaningful; images that should vary with the camera are not forced to agree. Against training on the same pairs without the consistency objective, orbital extrapolation improves in two seeds, interpolation does not, and an extreme compound shift sharply degrades one object suite. This is evidence for selective output consistency in simulation, not new-body adaptation. For touch, calibrated coordinates and compatible sensory layouts remain part of DexTouch-WM's transfer interface [67].

These comparisons locate adaptation at specific interfaces. The reviewed scene-transfer recipe retains a physical command interface while adapting visual experience; DreamZero fits the receiving body's policy, and DyPES-VLA retains body-specific action experts. SCVC instead retains action semantics while constraining selected outputs across views. Model selection should therefore identify which interface changes and which modules require fitting: shared visual experience supports transfer only through the command or decoding interface tested in each study. The SCVC regressions also make extrapolation and compound-shift tests necessary complements to gains on selected views. Section 3 follows how these predictions enter learning and action.

3 World Models in Physical AI Learning, Simulation, and Decision Making

Predictions can change command selection, policy parameters, action decoding, or correction after new observations. A simulator supplies interactions; an evaluator estimates behavior quality. We compare these operations rather than treating their benefits as interchangeable. This complements Hou et al.'s architectural taxonomy [28]: shared backbones and separate experts describe coupling, while the following sections track what consumes a prediction and when it changes physical action.

3.1 Planning with Predicted Consequences

Consequence planning must both propose candidate action sequences and estimate their future utility before committing to an action. These tasks require different operations: a predictive model estimates the consequences of sampled actions, a search procedure explores candidate sequences, and a task-dependent objective scores their predicted outcomes. PETS propagates state trajectories through probabilistic dynamics ensembles and uses the cross-entropy method (CEM) to search for high-reward action sequences [12]. With image observations, PlaNet infers a latent belief and applies CEM to predicted latent states and rewards, without rendering future images for scoring [24].

TD-MPC2 guides sample-based latent search with a learned policy prior and a terminal value [26]. These methods emphasize different components of the planning process—uncertainty-aware state propagation, inference of a compact planning state, and policy/value-guided search—rather than merely providing alternative forecast formats.

Goal-image control changes how predicted trajectories are scored. Visual Foresight predicts future images under supplied commands and evaluates candidates using designated-pixel or goal-image costs [14]. For designated-pixel tracking, its learned registration module updates the target object's pixel location from actual observations, rather than relying only on positions propagated through predicted frames. In its physical pushing comparison, registration-based tracking improves long-distance success, whereas short-distance success is comparable across the tested tracking variants. This distance-dependent result supports observation-based tracking for the reported long-distance pushing tasks; it does not isolate a general benefit for longer planning horizons or other manipulation settings.

A generic receding-horizon decision can be expressed as

$$a^*_{t:t+H-1} = \arg\max_{a_{t:t+H-1}} \mathbb{E}_{p_\theta}\left[\sum_{k=0}^{H-1} \gamma^k r(y_{t+k+1}, a_{t+k})\right], \tag{2}$$

with an initial action or chunk executed before replanning. Here y denotes the physical or learned predictive state, p_θ the predictive trajectory distribution, H the planning horizon, r the outcome score, and γ the discount factor. This template permits different optimizers and objectives, including terminal values, constraints, and uncertainty penalties; it does not identify predictor quality independently of search and scoring.

Two experiments locate different bottlenecks within this template. In action-conditioned JEPA planning, adding supervision from measured configuration and motion to inverse-dynamics training raises success by 2–22 percentage points across four simulated goal-conditioned tasks [55]. Both variants use latent-distance CEM, with 50 fixed start–goal problems per task and three independently trained seeds; physical measurements supervise training, not deployment. The result shows a more useful learned state for this planner without replacing its search rule.

The second experiment probes the candidate-selection side of the same planning template: before evaluating a scorer, one must establish whether the tested candidate pool contains a successful alternative. Yuan et al.'s restored-state WAM experiment records 16 candidate continuations at each of 48 states under a matched continuation, giving 768 branches [96]. Uniform selection yields 68.9% expected success and an outcome oracle 79.2%, but the entire gap comes from six mixed-outcome states; the other 42 offer no terminal-success gain from reranking under that continuation. Moreover, choosing wider-spread subsets of four or eight actions from 32 archived candidates increases action-space spread without increasing oracle success over the corresponding ordered subsets. This is stronger than simply asking for more diverse actions: in these saved pools, geometric spread does not identify additional successful choices. The oracle uses realized outcomes and is not a deployable selector. Its separate 64-state full-chunk panel and online $K = 4$ test use different pools and horizons. Section 4.4 examines practical scores and a proposed physical failure audit.

Taken together, the representation ablation improves the information used by a fixed planner, whereas the branching experiment measures the opportunity available within a fixed pool. Neither substitutes for the other. Longer rollouts can expose delayed effects, and more candidates can broaden search, but they also increase computation; subsequent observation feedback cannot undo an already executed chunk. The relevant comparison therefore includes candidate source, scoring rule, horizon, and the next opportunity to observe and revise.

Finally, the selected variable must be executable in the claimed setting. GeoWorld v1 combines a hyperbolic predictive representation with energy-based CEM over procedural action-label sequences between observed and goal clips [102]. Its reported improvements in annotated sequence success are more consistent in the video-clip-input evaluation than in the still-image-input evaluation. These are different evaluation protocols, not a controlled comparison of video versus image supervision. Moreover, the selected instructional labels are not robot motor commands: the results support sequence selection within the evaluated annotation protocols, not the motor grounding required by the physical planners above.

3.2 Driving and Navigation as an Interface Test

Driving and navigation expose a common design choice: prediction can compare supplied actions, provide context for generating an action, or support revision after feedback. The distinction determines what should be held fixed when attributing a gain to prediction.

For candidate selection, Drive-WM and NWM expose different search opportunities. Drive-WM generates multiview futures for three VAD command trajectories and selects with map/object rewards [84]. Its selected commands approach its ground-truth-command result, but adding map reward to object reward leaves average L2 at 0.80 m and changes collision from 0.27% to 0.26%. The overall selection gain and the marginal contribution of a reward component are therefore separate comparisons. NWM either reranks 16/32 fixed NoMaD proposals or constructs alternatives with CEM [6]. Switching between these modes changes candidate source and query allocation together. A selector's measured gain must consequently be interpreted against the opportunity supplied by its proposal mechanism.

Occupancy and next-state predictors shape action generation at different points. OccWorld jointly predicts occupancy and ego displacement from scene/ego history [107]; Drive-OccWorld ranks sampled paths with occupancy costs, refines the choice with BEV features, and conditions forecasting on that trajectory [93]. In Drive-OccWorld, ground-truth trajectory conditioning improves planning L2/collision relative to predicted trajectories, whereas predicted trajectories slightly improve occupancy/flow forecasting. Because model inputs and training also change, the contrast evaluates the coupled forecasting/planning interface. These contrasts distinguish jointly producing an action from scoring supplied trajectories; a change in prediction inputs or candidate source must be accounted for before assigning selection gains to forecast quality.

Predictive conditioning separates the training signal from information retained at execution. DriveVLA-W0 uses future-image objectives to train action features [49]; DriveDreamer-Policy retains ordered depth/video-query context while bypassing visual generation [112]. DriveWAM generates future latents before chunk decoding and retains selective video–action memory [75]. Its video-loss control tests supervision, while its memory control trades a small displacement-error increase on 20-second clips for lower cache use in separate 300-second profiling. That profiling measures memory cost, not reference-matched long-duration accuracy. These contrasts distinguish the cost of generating future information from the benefit of using it in the action decoder. NavThinker provides another form of runtime context: candidate-conditioned next-state features inform policy inference, while pedestrian predictions supply training rewards [29]. Its look-ahead ablation supports this input; physical Go2 deployment remains qualitative.

Before an action is issued, prediction can also revise the proposal itself. DAWN alternates action denoising with action-conditioned latent rollout [57]. Its predictor/refinement controls support this operation, although NAVSIM v2 aggregates remain below several listed baselines. Together with the supervision and context-readout comparisons above, this distinguishes a learned predictive objective, an input to the action decoder, and iterative refinement of its output.

Feedback determines the next opportunity to revise. ADriver-I describes predicted speed/steering to a video generator and feeds its frames into later steps [37]. Control comparisons use nuScenes and private highway data and video metrics use nuScenes; recurrent driving itself is qualitative self-generated video. NavWAM re-queries after actual chunk observations; its physical comparison supports the complete system, while predictive-head ablations are offline and future/value outputs optional [3]. DreamerNav instead trains its local actor through imagination and retains static global A^* guidance during execution [73]. NavForesee organizes waypoint generation through short-horizon geometry and long-horizon milestones [53]. Its controls support the milestones and language planner without isolating removal of the short-horizon input. These controls concern the context used for decisions; they do not by themselves establish a benefit from changing the observation or re-query schedule. Thus recurrence on generated frames, re-querying after observed execution, and retaining task-level context supply different evidence about the next decision.

These three comparisons require different controls. Candidate selection needs a specified pool, scoring rule and query budget; predictive action generation needs separate tests of training supervision, runtime context and pre-issue refinement. Feedback-based revision additionally needs an explicit observation schedule and action-commitment boundary. Evaluating each at its own interface makes the reported gain interpretable without treating a larger candidate bank, an additional predicted feature or a fresh observation as the same intervention. The driving and navigation settings in the top and middle panels of Table 2 retain the corresponding candidate and feedback protocols.

3.3 Policy Learning with Imagined Experience

What the optimizer consumes determines how imagination changes learning. Original Dreamer differentiates imagined returns through fixed latent dynamics; MBPO supplies short predicted state/reward transitions from real replay to SAC updates [23, 35]. DreamerV3 instead uses normalized REINFORCE for discrete and continuous actions [25]. PWM retains first-order return gradients and trains per-task actors from a shared multi-task model [19]. Its favorable MT30/MT80 results and poor single-policy MT30 results distinguish reusable dynamics from a reusable actor; model/training/optimizer choices also differ from TD-MPC2.

The horizon controls explain why these learning inputs are not interchangeable. MBPO's Hopper ablations retain much of the benefit with one-step branches, degrade with long branches, and outperform extra no-model SAC updates [35]. Original Dreamer's values extend returns beyond the finite rollout, while reward-only representation learning is insufficient in its visual-control tests [23]. Transition accumulation, return truncation, and state information are different weaknesses; these controls establish neither a common optimal horizon nor equal total compute.

For video-based VLAs, imagination can supply supervision or relative rewards without actor differentiation through video generation. VLAW adapts its simulator on successful and failed real rollouts, then uses real successes plus reward-filtered imagined successes as flow-matching targets [21]. WMPO adapts with target-policy interactions but converts binary rewards from imagined groups sharing initial frames into GRPO advantages [113]. Dynamic sampling discards and resamples all-success/all-failure groups. VLAW therefore selects successful demonstrations; WMPO needs within-group outcome variation. A stricter success label changes accepted targets in one optimizer and relative-reward groups in the other. The distinction locates model/reward error in the training quantity consumed, rather than treating every imagined rollout as the same additional experience.

VLAW's first-round drawing success is 22% for the base, 46% real-only, 50% synthetic-only, 60% real plus half the synthetic budget, and 68% for the full mixture [21]. After two rounds across five task categories, its pipeline reaches 0.868 versus filtered BC's 0.752 at matched real-rollout counts

Table 2. Where predictions enter decisions (§3): interfaces and selected *within-study* evidence.

System	Role	Setting / metric	Result / evidence
Driving			
Drive-WM [84]	Rank	nuScenes: L2 / collision	Random command → tree search: 1.02 → 0.80 m; 0.93 → 0.26%
OccWorld [107]	Joint	Forecast, 1–3 s	150 scenes
Drive-OccWorld [93]	Rank	nuScenes	Trajectory-conditioned planning/forecast
DriveVLA-W0 [49]	Train	Scale ablation	Future images shape action features
DriveDreamer-Policy [112]	Context	NAVSIM v1 PDMS	Action-only → depth+video: 88.0 → 89.2
DriveWAM [75]	Context	4-s ADE / FDE	0.83 / 2.47 (N = 1,000)
DAWN [57]	Refine	NAVSIM PDMS (N/P/R)	82.8 / 85.2 / 87.9
Navigation			
NavForesee [53]	Context	R2R-CE / RxR-CE	Milestone/planner; short input not isolated
NWM [6]	Rank	RECON (offline)	16 / 32 proposals
DreamerNav [73]	Train	Warehouse / Go2	1 sim.; 2 preliminary
NavThinker [29]	Context	Social-HM3D / Go2	Go2 qualitative
NavWAM [3]	Re-query	Physical success: N/O/W	19/24; 14/24; 4/24
World–action models			
UniPi [13]	Decode	Open-loop	Video → inverse dynamics
Fast-WAM [97]	Train/joint	RT / LB success (%)	RT: 91.8/90.6/91.3; LB: 97.6/98.5/98.0
V-JEPA Policy [101]	Context	LIBERO-Plus success	79.25% full; 68.81% no loss (1 seed)
LaRA-VLA T5 [4]	Joint	SimplerEnv (T/V/B)	64.6 / 63.5 / 68.8%
LaRA-VLA T9 [4]	Train/infer	SimplerEnv CoT (N/T/I)	55.2 / 61.4 / 68.7%
DreamZero [95]	Joint	Physical tasks	Joint denoising; refreshed K/V
JOPAT [20]	Joint	LIBERO-Long (L/T/J)	77.4 / 26.2 / 96.4%
Cosmos Policy [40]	Joint/rank	Physical tasks	Joint sample / candidate rank
Motus [9]	Joint	Robot adaptation	Supplied/generated actions

Role: Rank = score/search; Joint = world–action generation; Train = prediction-based training; Context = predictive input; Refine = pre-issue revision; Re-query = post-chunk observation; Decode = inverse dynamics. ADE/FDE = average/final displacement error. DAWN N/P/R = no predictor/predictor/refinement. NavWAM N/O/W = NavWAM/OmniVLA/NWM. Fast-WAM RT/LB = RoboTwin/LIBERO, with Base/Joint/IDM triples. LaRA T5 T/V/B = latent text/vision/both (Text-CoT off); T9 N/T/I = no CoT/train-only/train+inference CoT. JOPAT L/T/J = latent/track/joint (same DROID pretraining). V-JEPA retains future queries when future loss is removed. Text-only cells summarize mechanisms, not numeric outcomes; qualitative results are not success rates.

and 50 evaluations/task. Collection uses 50 real rollouts/task/round; generation uses 500/task before filtering. On a separate 40-trajectory reward audit, filtering reduces false positives from eight to two while increasing false negatives from seven to twelve. This is a label-selection tradeoff, not a policy-success comparison across thresholds. WMPO's separate physical evidence and training budget appear in Table 3; neither pipeline matches total generation, adaptation, and reward-computation costs.

World-VLA-Loop reconnects learning to physical experience by expanding the model-training dataset with policy rollouts [56]. Each model stage restarts from one ManiSkill-pretrained checkpoint and each RL stage from the base SFT policy: data accumulate, not continuously updated weights. Its simulator-evaluated learning curves, physical-policy comparison, and later two-iteration outcomes measure different stages. Replay anchoring, success filtering, and renewed collection thus change different learning inputs, not the current deployed action.

Table 3. How imagined experience trains policies (§3.3). Numeric entries are within-study outcomes, not a shared leaderboard.

Study	Imagined learning signal	Tasks / train / eval	Reported effect
Dreamer [23]	Latent return	20; 5 seeds	Full > reward-only
DreamerV3 [25]	Norm. REINFORCE	Multi-benchmark	Fixed HP (qual.)
PWM [19]	Shared WM / task actors	MT30/80; 10 seeds/task	Single MT30 weaker
MBPO [35]	SAC + short rollout	Hopper	1-step > long
VLAW [21]	Filtered successes	5 categories; eval: 50/task	VLAW / filtered BC: 86.8/75.2%
WMPO [113]	Binary group / GRPO	Base: 200 demos; DPO/WMPO: shared 128 real rollouts	53/60/70%; eval: $N = 30$/arm
World-VLA-Loop [56]	Imagined frames / rewards	30 physical rollouts*	Cup: 13.3→36.7%; cube: 26.7→40.0%

Notes: WM = world model; HP = hyperparameters; "qual." = no commensurate numeric value in this table. WMPO order: base/DPO/WMPO. VLAW compares two-round pipelines with matched *real* rollouts, not equal total generation cost. World-VLA-Loop compares SFT with RL post-training; *its caption states 30 physical rollouts without a verified per-cell allocation.

3.4 Memory, Failure Detection, and Recovery

Memory supports either prediction of persistent state or action generation from task history. Mem-World retrieves observations from action-conditioned, wrist-view-centered 4D memory for simulation [109]. Across five manipulation tasks, its predicted and physical policy success rates have Pearson correlation $r = 0.97$, compared with $r = 0.85$ for Ctrl-World. This task-level association evaluates policy assessment through simulation. WorldScape Policy 2.0 instead combines recent visual context, completed-subgoal events, and latent subgoal reasoning for action generation [76]. Its progress-aware policy context changes the acting system; the Mem-World correlation assesses the simulator as an evaluator. Neither endpoint measures recovery after a supplied failure.

Generated trajectories train different responses depending on which outcomes are retained. In Mem-World's separate post-training experiment, humans select 20–30 successful rollouts per task from 200 candidates; fitting the policy on these examples raises mean physical success from 58% to 72% across three long-horizon tasks [109]. This result evaluates the selection-and-post-training recipe, separately from the five-task evaluator correlation. Dream2Fix instead perturbs successful demonstrations inside Ctrl-World into four grasp-related failure types, verifies them, and trains a vision-language model (VLM) to output structured trajectory corrections [46]. Successful trajectories supply policy targets; synthesized failures supply correction targets. Dream2Fix's generated-test score measures partial-credit correction components, so physical recovery remains a separate endpoint.

Execution requires an intervention interface. Failure-Aware RL thresholds short-horizon latent constraint predictions to switch from a task policy to an offline-trained recovery policy [47]. Dream2Fix instead triggers VLM trajectory corrections from grasp keyframes under a fixed action budget [46]. ViFailback supplies a non-predictive comparison: an external VLM sends visual-symbol corrections through textual VLA prompts or point-based motion control, improving physical success after task-specific finetuning [98]. ViFailback's physical tasks are unseen in its dataset, but policy finetuning still uses expert demonstrations for those tasks. This identifies a useful correction interface rather than a wholly zero-shot or world-model-specific effect. These mechanisms separate avoidance, failure-conditioned recovery and overall completion—including episodes that never require intervention—as the response endpoints examined in Section 4.3.

3.5 Visual Plans and Joint World–Action Models

Predictive supervision shapes learned action features; runtime future information enters the current decision. UniPi generates task-conditioned video and decodes frame transitions through independently trained, task-specific inverse dynamics, executing open-loop [13]. Its future-to-command interface differs from querying consequences of supplied commands.

Fast-WAM tests this stage distinction within one model family [97]. Base action tokens read current-observation features, not future-video tokens. Removing video co-training preserves architecture and inference but costs 8.0 points on RoboTwin and 4.1 on LIBERO. Joint enables video–action attention and joint denoising; IDM generates future representations before actions and trains with noisy video tokens. These losses exceed the observed differences among the tested model configurations. These configurations change training and information flow, rather than switching inference alone.

Fast-WAM evaluates RoboTwin (2,500 clean and 25,000 randomized demonstrations; 30k steps, 100 trials/task/setting) and LIBERO (500 demonstrations per ten-task suite; 20k steps, 2,000 evaluations across 40 tasks) under distinct protocols [97]; Table 2 compares their simulation outcomes. Its Galaxea R1 Lite towel-folding experiment uses 60 demonstration hours and 30k steps. Removing video co-training lowers physical success to 10%. Within the model family, IDM leads physical success at 810 ms inference latency, whereas Base runs at 190 ms and completes tasks faster; pretrained $\pi_{0.5}$ leads both success and completion time. Thus video supervision and runtime future-token generation have different measured benefits and costs.

V-JEPA Policy's action expert reads context-position keys and values (K/V) updated by learnable future queries, not final predicted latents [101]. Observation, instruction and current proprioception condition this path; candidate robot commands do not query it. Removing future loss while retaining queries tests supervision with runtime context intact. A deployment comparison would instead vary context access through a training-compatible action interface. Thus omitting rendered video does not remove predictive-context access (Figure 4).

LaRA-VLA's action expert attends to observations, instructions, latent reasoning and future visual latents [4]. Its Table 5 tests supervision combinations, favoring joint latent text/vision over either alone. Table 9 separately tests CoT during training and inference, favoring retention over training-only use. Removing inference CoT removes reasoning and image-next tokens together; the nonuniform task effects therefore concern the joint bundle, not a visual-only intervention.

Joint generation retains future information inside action sampling. DreamZero couples video and action denoising and replaces generated-frame K/V entries with observations after chunk execution [95]. JOPAT adds 2D motion tracks and visibility prediction [20]. With DROID pretraining, its comparison matches pretraining data and downstream schedules but changes supervision, branch structure and runtime information together, supporting the combined appearance/motion design. All variants use the same action-free video pretraining setup. Executing decoded actions alone leaves the future tokens used during sampling intact.

Query mode further distinguishes joint generation from candidate evaluation. Cosmos Policy directly samples actions, futures and values, or scores candidate-conditioned futures after policy-rollout adaptation [40]. Motus uses flow-derived latent actions weakly aligned to commands: world-model mode accepts actions, while policy/joint modes generate them [9]. The required action query therefore determines which mode to evaluate; Section 4.3 examines feedback during execution.

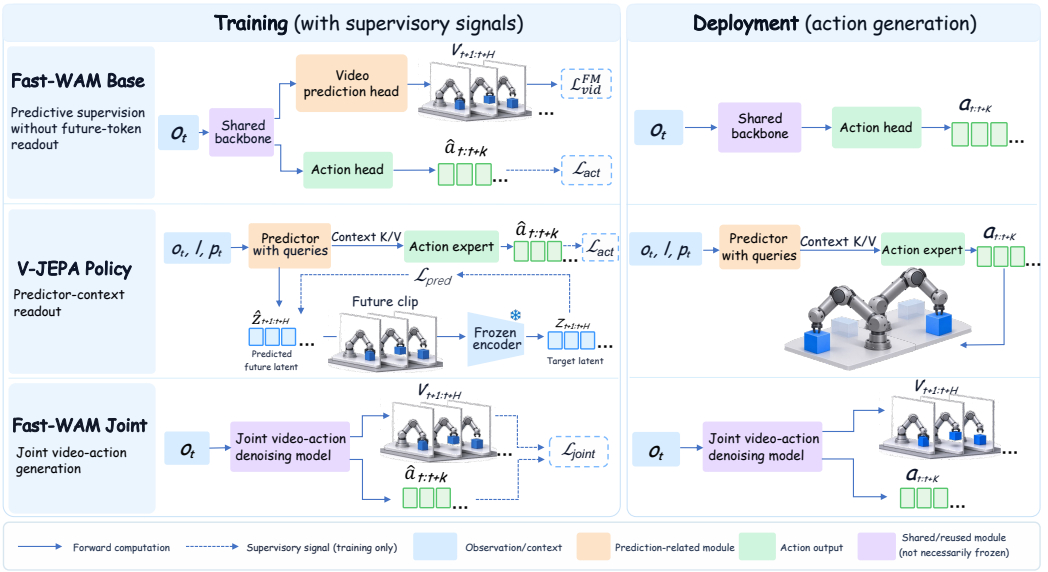


Fig. 4. Training and deployment information paths in Fast-WAM Base, V-JEPA Policy, and Fast-WAM Joint [97, 101]. Base and Joint are configurations of one study. V-JEPA Policy uses future-latent supervision during training and predictor context during action generation.

3.6 Reasoner-Assisted Action Revision: A Boundary Comparator

The predictive interfaces above shape action generation or selection through modeled future information. External reasoners form a useful boundary case: they can accept, edit or replace a proposed command without querying its predicted environmental consequences. These systems provide a boundary comparator for the role of predicted consequences in action revision. Su et al. report 48% success for Astra revising $\pi_{0.5}$ proposals versus 26% for direct Astra on ten selected RoboDojo tasks (five aligned cases per task) [77]. This comparison changes the proposal source and review workflow together, so it does not isolate the reviewer's effect or establish a forward-model benefit. The later Galbot report reuses the same manipulation result rather than independently replicating it; locomotion reasoning also pauses physics [16]. A separate direct-policy evaluation reports 22.48% across 42 tasks and 2,100 trials under a different protocol [100].

FLUX 3 Action provides accept, edit, replace, and stop decisions while tracking time and inference cost [10]. These operations demonstrate a command-revision interface, not prediction-specific benefit. The relevant distinction is whether predicted action consequences inform revision, not simply whether a reasoning model is involved.

Across this chapter, prediction can change policy training, candidate scoring, action decoding, or correction of still-executable actions. These operations require distinct evidence: trained-policy success, realized outcomes within tested candidate sets, or successful intervention. Section 4 examines the corresponding endpoints.

4 Evaluation, Reliability, and Research Directions

A prediction can be accurate, useful for a particular decision, and reliable during execution to different degrees. We compare the tests that distinguish these properties and then examine the explanations left unresolved by their controls.

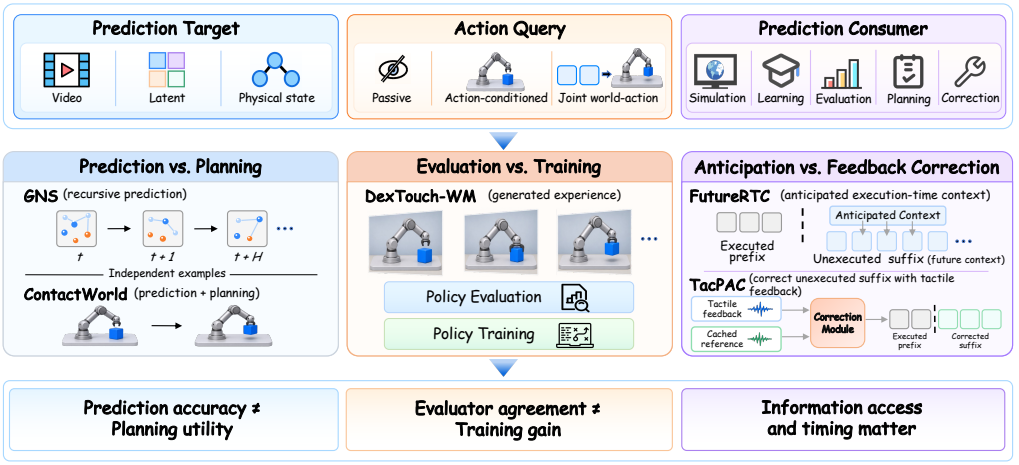


Fig. 5. From world-model design to claim-matched evidence. Three contrasts distinguish recursive fidelity from planning utility, evaluator agreement from training outcomes, and execution-time anticipation from correction. GNS/ContactWorld and FutureRTC/TacPAC are independent studies rather than matched experiments.

4.1 Prediction Quality and Decision Utility

Evaluation requires a separate map of what a result establishes. We distinguish six *evidence dimensions*, adapted from evaluation questions developed across prior world-model and embodied-AI surveys [28, 36, 94]. *L1, perceptual fidelity*, tests visual coherence; *L2, state and physical fidelity*, tests task-relevant dynamics, geometry, or contact state. *L3, action realization*, checks whether changing a supplied command produces the corresponding robot or agent motion, while *L4, interaction grounding*, checks whether the *realized motion* produces the corresponding environmental response, including object displacement or contact. This distinction follows the chain *commanded action* → *realized motion* → *environment response* → *new observation*. *L5, planning and policy utility*, measures effects on action choice or learning under a defined consumer and protocol; *L6, operational reliability and safety*, measures failures, intervention timing, recovery, and constraints during execution. These are claim-dependent checks, not six mandatory sequential gates or a single aggregate score: an L1 or L2 gain does not establish an L5 benefit, and a policy-evaluation correlation is not a newly trained policy's success rate.

Figure 5 links the three design axes in Section 1.2 to these evidence dimensions. The design axes specify what is available to a consumer; the changed operation, endpoint, and controlled alternatives determine what a reported result can support. The three findings in Section 1.1 show why that distinction matters without claiming a universal causal ranking.

A model-selection question is whether a benchmark measures the computation that will consume the prediction. AVID makes this distinction within one evaluation: RT-1 frame metrics assess generated appearance, whereas Action Error Ratio (AER) compares generated and real videos through a video-to-action regressor; AER is a motion proxy, not a robot-execution measure. Its v2 test uses 1,024 held-out 16-step trajectories [70]. GNS instead intervenes on recursive inputs (Fig. 4(g,h) of the original study): intermediate noise gives the best rollout GOOP result, while increasing noise worsens clean one-step fit; the reported figure gives medians and quartiles across seeds and test sequences but not an exact seed count [71]. These controls support a specific choice: appearance metrics cannot substitute for motion-sensitive tests, and a simulator trained

for recursive use may trade one-step fit for rollout robustness. Neither result alone establishes downstream policy benefit.

ContactWorld tests whether a predicted contact representation changes planning outcomes, and its tables answer distinct questions [104]. Table VI reports medians over 12 tasks: PointCloud+TacRGB has the lowest six-step relative error growth (3.85; fitted growth rate 0.295), a model-dynamics diagnostic rather than a planning score. In Table III, PointCloud+TacFF leads the 12-task simulated-planning mean (36.1% versus 32.1% for point clouds); each task–modality configuration averages 100 rollouts at each of four goal offsets, or 400 rollouts. Table VIII isolates four tasks: point clouds score 32.3%, full TacFF 39.5%, visual-only terminal cost 38.6%, and dimension-matched latent width 36.5%. These are means and standard deviations across three seeds and four offsets, with 100 rollouts per seed–offset pair (1,200 per task–method). The controls make a cost-only or width-only explanation insufficient for the aggregate result, but do not isolate a universal tactile-prediction effect: on Valve, full TacFF scores 60.8%, visual-only 53.3%, and point clouds 53.6%; on Dim exploration, dimension matching falls to 9.5% versus 14.9% for point clouds. Encoders and goal metrics are learned separately, so their geometry remains coupled to the predicted representation (Case B).

Physical-event benchmarks should not be read as action-response evaluations. IntPhys 2 tests judgments about possible and impossible video events; Physion evaluates predicted object contact across eight scenarios, with 150 test stimuli per scenario; PHYRE measures placement-solving under deterministic folds and a budget of up to 100 attempts per task [5, 7, 11]. Their units and endpoints differ, and none by itself establishes that a commanded action produces the predicted movement. For the data and representation interfaces reviewed in Sections 2.1–2.3, a paired-state intervention varying command direction, magnitude or duration and separately measuring agent motion and object/contact response would add this missing endpoint. This intervention is proposed for future evaluation.

Policy-evaluation results depend on what serves as the reference. WorldArena compares five $\pi_{0.5}$ policies with RoboTwin; its proxy metrics can correlate with performance while overestimating success, and its evaluator rollout count is not specified in the reported section [72]. WorldGym reuses recorded physical starts for three policies across 17 tasks, with ten starts per policy–task and uncertainty summarized across tasks; it therefore tests consistency on sampled physical initial states, not fresh closed-loop physical execution [68]. WorldEval reports 40 rollouts for each of five physical tasks and policy-level agreement, but does not specify the allocation among policy–task cells [50]. In driving, WorldLens measures the response of a fixed planner to generated observations, separately from route completion under model feedback [51]. These protocols can diagnose evaluator agreement or sensitivity of a fixed consumer; they do not establish that a newly trained policy will benefit.

IWS evaluates policy scores on four tasks, each using 20 matched initial configurations sampled from the simulator's training distribution (Fig. 7 in the original IWS paper) [85]. Its separate data-mixture experiment (Fig. 5 in the original IWS paper) trains each policy on 100 demonstrations per mixture, with 100 MuJoCo or 10 physical evaluation trials per reported point. The reported mean normalized task scores change from simulator-only to real-only data by learner: $\pi_{0.5}$, 73.1% to 88.8%; DP, 87.9% to 90.3%; and ACT, 76.2% to 73.6%. These are task scores rather than a uniform binary-success endpoint. Agreement on matched evaluation starts does not determine which data mixture works best for a retrained policy.

DexTouch-WM compares evaluator agreement and training utility in separate experiments (Figure 6) [67]. Table III computes each task's Pearson correlation across three evaluated policies; its four-task mean is 0.844 for WM-Mix and 0.646 for WM-Robot. Table IV instead retrains each learner on 100 real plus 100 imagined trajectories, against a 200-real reference. Recomputed four-task

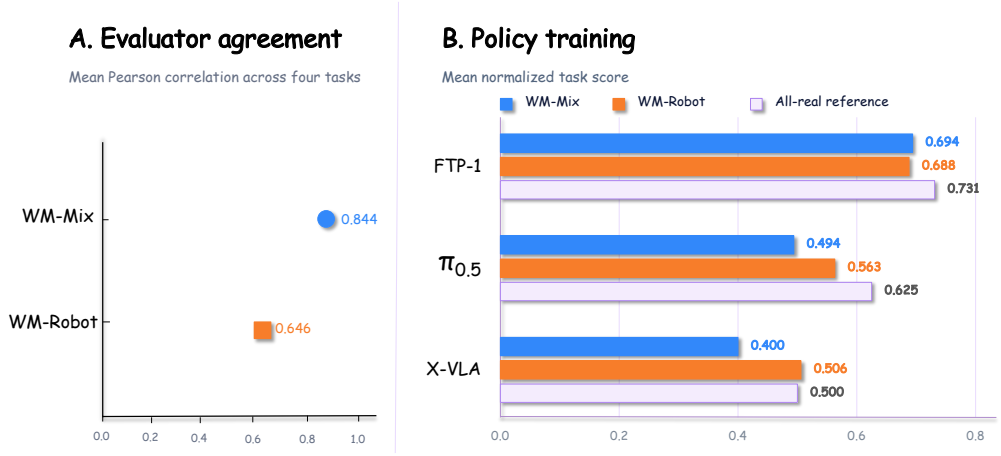


Fig. 6. Evaluator agreement and policy-training outcomes from separate DexTouch-WM experiments [67]. Panel (a) shows four-task mean evaluator Pearson correlations from Table III; panel (b) plots four-task mean normalized training scores computed from Table IV. Imagined-data conditions use 100 real plus 100 imagined trajectories, and the all-real reference uses 200 real trajectories. Small learner-level differences have no reported uncertainty test.

Table 4. What is evaluated (§4.1). The marker identifies a policy-evaluation protocol.

Study	Pol.?	Endpoint	Evaluation N / unit	Key result / limit
Prediction or physical-event tests				
AVID [70]	×	Frame / AER	1,024 trajectories ×16 steps	Regressor proxy
GNS [71]	×	1-step / rollout	GOOP; repeats NR	No planner
IntPhys 2 [11]	×	Event judgment	1,012 / 344 videos	Paired/single
Physion [7]	×	Contact event	8 × 150 stimuli	No motor execution
Policy evaluation or ranking				
WorldArena [72]	✓	VLM judgment	5 $\pi_{0.5}$; N NR	Proxy optimism
WorldEval [50]	✓	Video verifier	5 tasks; 40 rollouts/task	Policy–task N NR
WorldGym [68]	✓	Rubric	3 policies; 17 tasks; 10 starts	Starts reused
IWS [85]	✓	Score agreement	4 × 20 starts	Mixture separate
DexTouch-WM [67]	✓	Pearson r	4 tasks; 3 policies/task	Mix/Robot .844/.646

Legend: Pol.? ✓ = policy outcome/ranking; × = prediction/event test. AER = action error ratio; NR = not reported. IWS initial states match the simulator training distribution; its mixture training is a separate experiment. DexTouch additionally uses 10 matched rollouts per policy–task–environment and five raters.

means (Mix versus Robot) are 0.694 versus 0.688 for FTP-1, 0.494 versus 0.563 for $\pi_{0.5}$, and 0.400 versus 0.506 for X-VLA. Robot leads on two learners, whereas FTP-1 weakly favors Mix; on X-VLA's Stand Bottle task, Mix scores 0.000 versus 0.600 for both Robot and all-real data. Excluding that task in a post-hoc sensitivity calculation reverses the X-VLA ordering: Mix averages 0.533 across the remaining three tasks versus 0.475 for Robot, compared with 0.400 versus 0.506 across all four. The reversal shows that the X-VLA aggregate is sensitive to task composition. Neither the 0.006 FTP-1 difference nor Robot's 0.006 excess over the X-VLA all-real mean (0.506 versus 0.500) has a reported uncertainty test. Thus evaluator correlation does not select a learner-independent training generator, nor do the separate protocols isolate a generator-only causal effect. VLAW addresses another policy-training endpoint (Section 3.3) [21].

4.2 Uncertainty, Constraints, and Runtime Guardrails

Safety mechanisms intervene at different operations: exploration, policy optimization, command acceptance or alarm generation [18]. Dynamics confidence, task-failure risk and physical-harm probability consequently require different validation.

Classical certificates establish a response under explicit assumptions. Confidence-bounded Lyapunov learning restricts exploration to an invariant region; predictive safety filtering tests a backup trajectory and minimally modifies the proposed input; a control-barrier-function QP constrains its instantaneous derivative [1, 8, 81]. Their guarantees require a valid state/uncertainty model and, respectively, an initially stabilizing policy, feasible terminal backup or admissible safe input. A visual forecast supplies none of these physical interfaces automatically.

Learned penalties and model-based planning address different failure modes. SafeDreamer compares online search with background safe-actor learning; on SafetyPointPush1, the background actor discovers a pushing strategy that finite-horizon, finite-candidate online search misses, showing that model access does not remove a search-horizon bottleneck [32]. SMBPO's formal result assumes calibrated set-valued dynamics, a specified unsafe set, bounded time from irrecoverability to violation, and a sufficiently large penalty; its Bellmin operator takes the minimum over possible successor states [80]. SMBPO's continuous-control implementation instead fits critic targets averaged across ensemble-model rollouts, which the authors found less conservative in practice. The Bellmin-based theorem therefore does not certify the experimental update, nor does either study establish a general safety guarantee for learned visual world models.

An alarm also differs from a completed intervention. Foresight calibrates a trajectory-wide alarm on action-conditioned latent scores, selecting α by balanced accuracy in three-fold cross-validation; the remaining folds use a 6:1:1 training/validation/calibration split [99]. Under its exchangeability condition, the guarantee bounds alarms on successful rollouts rather than certifying a physical cost or executable backup. Its 404 physical evaluations combine 360 policy–task trials (three policies, three tasks, 40 trials each) with 44 separate Franka episodes; this heterogeneous total is neither one independent calibration cohort nor the calibration-set size. DreamAvoid ranks candidate videos and replaces a chunk while high-level command publication pauses and the low-level robot holds position; Recovery RL switches to a learned recovery policy, with physical tests of its model-free variant [15, 79]. The first is a calibrated alarm, the second a candidate-based action replacement, and the third a policy switch. They establish different links in a response chain (Table 5); none should be reported as the same endpoint.

Table 5. Safety claims, alarms and executed interventions (§4.2–4.3). A formal guarantee and an empirical improvement are different endpoints.

Study	Thm.?	Mechanism	Endpoint / condition	Scope / N
Formal or conditional guarantees				
Berkenkamp [8]	✓	Confidence + Lyapunov	$1-\delta$ invariance	Safe initial policy
Safety filter [81]	✓	Backup feasibility	Constraint feasible	Terminal / tightening
CBF-QP [1]	✓	Barrier / QP	Set invariant	Control-affine
SMBPO [80]	✓	Set-valued dynamics	Bounded Bellmin	Emp. update ≠ proof
Reported interventions and measurements				
SafeDreamer [32]	–	Reward / cost WM	Return / violation	5 runs; 10 × 1,000
Foresight [99]	–	Risk / alarm	AUC / BalAcc	404 = 360 + 44
DreamAvoid [15]	–	Ranked action chunk	Physical failure	4 × 40/arm
Recovery RL [79]	–	Risk-critic switch	Success / violation	Model-free physical

Legend: Thm.? ✓ = formal statement under cited assumptions; – = no guarantee asserted *by this table.* BalAcc = balanced accuracy; AUC = area under the curve. SMBPO's Bellmin theorem does not certify its ensemble-averaged update. Foresight's 360 and 44 episodes are heterogeneous; alarm is not correction. DreamAvoid does not observe rejected outcomes.

4.3 Operational Reliability and Attribution

A runtime response requires information to arrive before the relevant action becomes committed. Astra pauses simulated physics during locomotion evaluation, whereas DreamAvoid pauses high-level command publication while the low-level hardware holds position [15, 16]. These controls manage elapsed time differently; neither pause should be treated as evidence of zero inference latency in an unconstrained deployment.

FutureRTC's Table 3 isolates action-index alignment while holding an execution-time oracle context fixed: using the first generated action yields 96.6% success, whereas starting at index $d = 20$ after the same delay yields 0.9% [38]. This is an oracle-context diagnostic, not evidence that the model can access that future observation when it generates the action; the table does not report N or seed count. TacPAC instead changes access to cached predictive tactile keys while retaining the plan, current tactile input, expert and parameter count. With 20 physical trials per method on each of five tasks, access to the cached tactile reference raises the equally weighted mean of the five task success rates from 47% to 64% [59]. Its contrast supports a benefit from the reference-access operation under that setup, not from prediction in isolation or a general comparison with online planning.

TacPAC's execution trace reports that 1.6% of expert calls overlap robot advancement by at least one control step; the denominator is calls, and the paper does not state the call count. Asynchronous updates alter only the unexecuted action suffix. The reported mean computation times are 628.6 ms for ten-step action-chunk generation, 62.6 ms for one clean pass that fills the reusable cache, and 30.4 ms for one cache-based correction from incoming touch [59]. Prefill is performed once per chunk and can overlap initial execution; these stage timings do not measure end-to-end sensor-to-actuator latency or its tail. The evidence therefore distinguishes access to a predictive reference, action-index alignment and update timing for an unexecuted suffix (Figure 7), without reducing the two mechanisms to one runtime-efficiency claim.

Invocation frequency is a different computation tradeoff. GeoBoN's simulated gated sampling uses 24.7% of decisions, obtaining 67.9% success at 1.29 s average latency versus 68.4% at 3.65 s always-on [105]. KnowNo instead changes who supplies information by asking a human when a calibrated language-planning set contains multiple candidates [69]. Neither average model latency nor semantic uncertainty is a physical response guarantee.

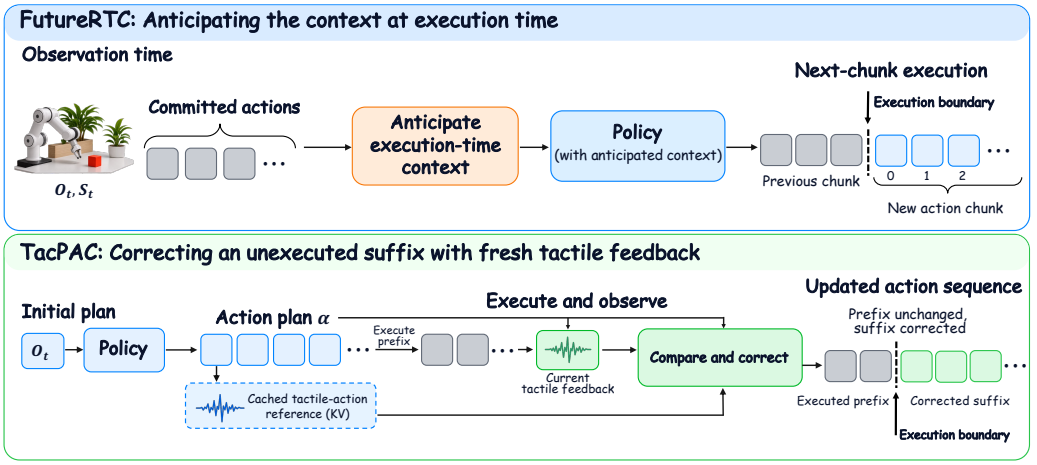


Fig. 7. FutureRTC anticipates execution-time context; TacPAC combines fresh tactile feedback with cached predictive context to correct an unexecuted action suffix [38, 59]. The executed prefix is unchanged. The panels illustrate distinct mechanisms on schematic, not comparable, timelines.

Perturbation studies localize only the measured stages. Lee and Zhang's 50 simulated drawer tasks show stage-dependent orderings under ten sensing degradations, but primary paired success intervals include zero and the restoration probe executes no recovered trajectory [45]. WAM/VLA perturbation rankings also change across RoboTwin and LIBERO protocols with unmatched training configurations [103]. These profiles identify tests worth separating, rather than a causal architecture ranking.

Response outcomes extend from detection to continuation. Foresight and FoMo-FD measure alarms; TacPAC executes corrections; Dream2Fix measures correction after failed physical attempts [33, 46, 59, 99]. LIBERO-Recover's gains on retry/adaptation/restoration do not consistently improve nominal success [54]. REBOOT supplies the missing continuation supervision through 2,160 demonstrations across 18 assembly tasks, equally divided between expert and human recovery, rather than establishing a deployed autonomous recovery rate [63].

Intervention can itself cause failure. On two physical Franka tasks, VLA-FixBench's automatic judgers achieve 0–25% completion versus 30% policy-only and 65% human intervention [91]. AgentChord's precompiled failure branches give 77.5% versus 72.5% for its strongest listed comparator on six physical tasks, 20 trials each [90]; this is final completion, not conditional recovery. Reliability therefore requires measuring the consequence as well as the timing of an intervention.

4.4 Research Directions and Synthesis

Across the five consumers, the experiments connect predictive utility to the quantity used by the downstream computation. Learned simulation requires robustness when predicted states become later inputs, as GNS's perturbation control demonstrates [71]. Policy learning depends on the optimizer's input: Dreamer and MBPO consume imagined returns or transitions, while VLAW and World-VLA-Loop use generated experience and renewed physical data [21, 23, 35, 56]. IWS and DexTouch-WM further show why evaluator agreement and generated-data utility require separate assessment for the intended learner [67, 85]. For planning, ContactWorld couples predicted evolution to learned goal distance, while Drive-WM and NWM expose candidate-dependent selection interfaces [6, 84, 104]. For execution-time intervention, FutureRTC tests context/index alignment

and TacPAC tests predictive-reference access during suffix correction [38, 59]. These comparisons identify the outcome needed to validate each use: recursive error, trained-policy performance, evaluator agreement, selected-action outcome or effective intervention.

Our synthesis connects consumer-specific outcomes to reported controls and adverse outcomes, building on prior surveys' evaluation principles [28, 36, 58, 74, 82, 94]. The L1–L6 dimensions in Section 4.1 identify what each endpoint measures. The original experiments test different interventions and protocols, so their outcomes are not pooled effects. Cases A–C distinguish these reported observations from proposed, unexecuted controls for candidate availability and scoring, predictive dynamics and learned goal geometry, and prediction context and correction timing.

Case A: candidate availability, prediction and selection. Reported evidence. The same-state experiment in Section 3.1 tests 16 candidate actions at each of 48 restored states under the same continuation policy [96]. A *mixed-outcome state* has both successful and failed continuations in its tested pool. Only six states meet this condition and can gain terminal task success by selecting a successful branch instead of a failed one; this is a retrospective outcome-oracle opportunity, not a realized scorer gain. Wider-spread action subsets do not raise oracle success in those pools. GeoBoN measures a different offline proxy: average displacement error (ADE), the mean trajectory-position deviation from a reference, enters its oracle-normalized error recovery (ER), $\mathrm{ER} = (\mathrm{ADE}_{\mathrm{first}} - \mathrm{ADE}_{\mathrm{selected}})/(\mathrm{ADE}_{\mathrm{first}} - \mathrm{ADE}_{\mathrm{oracle}}) \times 100\%$. On fixed eight-candidate RoboCasa/X-WAM dumps, consensus attains 8.6% ER, yet its online success is 77.4% versus 80.8% for the base policy [105]. Thus a positive offline trajectory-error recovery does not guarantee better executed success; neither experiment reveals outcomes for untested alternatives.

DreamAvoid's DA-ABL physical audit starts from 44 failures among 160 trials (40 trials on each of four tasks). Two cases lacked a timely intervention and were excluded; among the remaining 42 scored failure-causing chunks, 36 were below the diagnostic threshold but still executed because they ranked highest [15]. This documents a limitation of rank-only acceptance, not whether another safe candidate existed: rejected candidates' physical outcomes are unobserved, so the audit does not show that the selected chunk was misranked relative to a viable alternative. GeoBoN's false low-score selection rate rises from 7% with two candidates to 32% with sixteen, a separate proxy result that also prevents assuming a larger pool alone will fix selection. *Unresolved attribution.* Candidate availability, forecast quality and scoring remain competing explanations.

Proposed control (not executed). A conditional extension would branch reproducible physical failure states while fixing candidate set, continuation policy, horizon and stochastic repeats. Success of a rejected candidate would establish selection regret for that observed pool; failure of every tested candidate would not establish that the pool was exhaustive. Crossing predicted and realized consequences with a validated score could then separate forecast error from valuation error. The test requires valid resets and compatible inputs. Its outcome would determine whether the observed failures warrant improving selection within the tested pool or investigating alternatives beyond it.

Case B: predictive representation and goal-dependent scoring. Reported evidence. GNS's training-input perturbations trade clean one-step fit for recursive accuracy [71]. ContactWorld instead finds different representation preferences for prediction-error growth and goal-scored planning [104]. As detailed in Section 4.1, its four-task visual-cost and dimension-matched controls retain aggregate gains over PointCloud-only, but the visual-cost control lowers Valve success and dimension matching lowers Dim success below PointCloud-only. Neither the tactile term in the planning cost nor increased latent dimensionality alone accounts for the aggregate multimodal gain. *Unresolved attribution.* The controls do not independently separate predictive representations from goal geometry: visual-cost scoring retains visual–tactile prediction, while dimension matching retains both modalities.

Proposed control (not executed). A discriminating follow-up would freeze a compatible encoder, target and goal metric; match sensing, capacity, data and search; and vary predictive contact input separately from scoring access. Predictor-only gains would weaken a scoring-only account, while scorer-only or joint-only gains would implicate valuation or interaction. Missing-modality controls should be retrained to avoid out-of-distribution masking. A common target must preserve the task distinctions needed by both the predictor and the goal metric.

Case C: predictive computation, feedback and executable action windows. Reported evidence. Training and deployment require separate comparisons. Fast-WAM removes video supervision while preserving its Base action readout, testing how predictive training changes the learned policy [97]. LaRA-VLA separately tests supervision combinations and retained inference information, with visual and reasoning tokens coupled (Section 3.5) [4]. These are distinct interventions even when both improve a final policy score.

At deployment, DAWN refines an action hypothesis before commitment, while NavWAM acquires observations after an executed chunk [3, 57]. FutureRTC tests context/index alignment using an online-unavailable oracle condition; TacPAC tests cached predictive-reference access and correction of the unexecuted suffix (Section 4.3). These results concern information available at different times and through different interfaces. *Unresolved attribution.* They do not constitute one shared experiment on online correction, so the role of prediction, feedback and information timing cannot be separately inferred across these studies.

Proposed decoupled controls (not executed). A training-stage comparison would hold architecture, data protocol and deployment path fixed while varying the predictive loss or target, measuring the resulting policy outcome separately from deployment-time access. A deployment-stage comparison would instead fix the trained policy, committed prefix and available sensor stream while varying predictive context, reference access or re-query timing. Candidate methods should share a specified computational budget (e.g., predictor calls per decision or accelerator time) and report wall-clock latency, missed action deadlines and task outcomes separately. Context inputs must remain training-compatible and preserve execution indices; online-unavailable oracle observations are idealized diagnostic references, not proven upper or lower bounds on success. Feedback cannot rewrite actions already committed.

Conclusion. World-model selection depends on the computation that consumes a prediction. A simulator requires recursive-rollout validation; a planner additionally requires a useful scoring representation and executable alternatives. Policy-training generators must be tested with the intended learner and data recipe, while runtime interventions require context and feedback aligned with actions that remain editable. The reviewed controls and adverse outcomes make these choices concrete across simulation, manipulation, driving and navigation. Cases A–C extend this synthesis into proposed tests of candidate availability, learned goal geometry and execution-time information, providing a focused validation agenda for the mechanisms that existing comparisons leave unresolved.

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